Jacy

Boston, MA

SUMMARY STATEMENT

Master's candidate in Data Analytics Engineering at Northeastern University with 10 years of experience as a Licensed Architect leading large-scale design and planning projects across diverse sectors. Skilled in translating ambiguous problems into structured, data-driven solutions; adept at stakeholder communication, project ownership, and delivering results under constraints.

EDUCATION

Northeastern University (Boston, MA) — Expected May 2027
Master of Science, Data Analytics Engineering — GPA: 3.98/4.0
Relevant Coursework: Data Management (SQL, NoSQL), Data Mining & Machine Learning, Data Visualization & Computation (Python), Statistics & Probability, Human Factors

Nanyang University (Nanyang, China) — June 2013
Bachelor of Engineering, Architecture

TECHNICAL SKILLS

- **Programming:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL, NoSQL, Bash, Data Structures & Algorithms
- **Databases:** MySQL, PostgreSQL, MongoDB, Neo4j/Cypher; schema design, query optimization, ETL pipelines
- **Machine Learning & Statistics**: Regression, Classification, Clustering, Time-Series, Dimensionality Reduction, Feature Engineering, Model Evaluation
- **Data Visualization:** Plotly, Altair, Seaborn, Matplotlib, Tableau, Power BI
- **AI & LLM:** Anthropic Claude API, Prompt Engineering, LLM application development, Streamlit
- **Cloud & Tools:** AWS, Git/GitHub, Jupyter, VS Code
- **Domain Expertise:** Built environment analytics, spatial data, BIM, AutoCAD, SketchUp, Revit, Adobe Suite (Photoshop, Illustrator, InDesign)

RELEVANT PROJECTS

Job Application Assistant — AI-Powered Résumé Tailoring Tool — Summer 2026
Personal Project | Python, Anthropic Claude API, Streamlit | https://github.com/Jie-Hsu-D/job-application-assistant.git

- Built a Python web app (Claude API + Streamlit) that rewrites raw experience into tailored, job-specific résumé bullets, with anti-hallucination guardrails constraining the model to reformulate only user-provided facts.
- Designed a structured system/user prompt architecture and robust API response parsing; managed secrets with python-dotenv and published on GitHub.

Spatio-Temporal Activity Pattern Classification — ML Pipeline on Urban Dataset — Summer 2026
Northeastern University | Python, Pandas, NumPy, scikit-learn, Matplotlib

- Built end-to-end classification pipeline on 2.75 GB spatio-temporal dataset (grid-cell × hourly resolution) to model urban activity patterns across weather and temporal dimensions.
- Engineered features from 5 correlated activity variables (r=0.85-0.98) via dimensionality reduction; retained independent weather/temporal features to avoid multicollinearity.
- Benchmarked 5 classifiers with ROC/AUC and confusion matrices, achieving ~90% balanced accuracy; documented reproducible workflow and evaluation.

Boston Restaurant Database - End-to-End Data Pipeline — Spring 2026
Northeastern University | MySQL, MongoDB, Python | https://github.com/Jie-Hsu-D/Boston-Restaurant-Database.git

- Designed and implemented a Boston-area restaurant database to support community sharing and dining insights, primarily using relational MySQL and demonstrating NoSQL concepts (e.g., MongoDB playground)
- Developed conceptual data models using EER and UML, mapped them to relational schemas, and implemented the databases in SQL using DDL and DML
- Created SQL queries and integrated the database with Python applications to retrieve and analyze data for insight generation
- Strengthened end-to-end data management skills, including problem definition, requirements analysis, database design, implementation, and data-driven application development

U.S. Food Import Data Pipeline & Analysis — Spring 2026
Northeastern University | Python, Pandas, Matplotlib | https://github.com/Jie-Hsu-D/U.S.-Food-Import-Analysis.git

- Analyzed a dataset of 19,800+ records spanning 25 years of U.S. food import data using Python (Pandas, Matplotlib)
- Performed end-to-end data analysis, including data cleaning, transformation, and exploratory analysis
- Selected appropriate visualization techniques (e.g., time series, category comparisons, distributions) to translate complex data into clear, structured insights
- Automated data processing and visualization workflows, improving efficiency and enabling scalable analysis